



Via Electronic Submission

National Science Foundation
Networking and Information Technology Research and Development (NITRD) National
Coordination Office (NCO)
2415 Eisenhower Avenue
Alexandria, VA 22314

Subject: Response to the Request for Information on the Development of an Artificial
Intelligence (AI) Action Plan

COMMENTS OF CORNING INCORPORATED

I. INTRODUCTION AND SUMMARY.

Corning Incorporated (“Corning”) respectfully submits the following comments in response to the Networking and Information Research and Development, National Coordination Office (“NITRD NCO”), the Office of Science and Technology Policy (“OSTP”), and their Request for Information (“RFI”) on the development of an Artificial Intelligence (“AI”) Action Plan. As mentioned in the RFI, this Plan, as directed by a Presidential Executive Order on January 23, 2025, will define the priority policy actions needed to sustain and enhance America's AI dominance, and to ensure that unnecessarily burdensome requirements do not hamper private sector AI innovation.¹

Corning applauds the opportunity provided by the NITRD NCO and OSTP to highlight with this RFI the essential role that data centers have—and will continue to play—in powering the development of critical and emerging technologies like AI. Accelerated by the astonishing growth in AI over the last several years, domestic and global demand for data is significantly increasing, and the amount of bandwidth required to meet this growth will only continue to grow. As such, data centers will need to produce exponentially more processing power and transmit more data faster as new technologies are developed and gain increased adoption. Given the significant computing demands that come with AI and other emerging technologies, Corning is confident that the forthcoming AI Action Plan will be a valuable resource in ensuring that the data center industry can continue to “yield substantial benefits” for the United States both economically and socially.

For more than 170 years, Corning has applied its unparalleled expertise in specialty glass, ceramics, and optical physics to develop products that have created new industries and transformed people’s lives. Most notably, Corning is the inventor and an industry-leading supplier of optical fiber that has brought high-speed broadband Internet access to millions of consumers nationwide. Headquartered in Corning, New York, Corning continues to be a leading innovator in the development of communications infrastructure. As a testament to our

¹ See *Executive Order, Removing Barriers to American Leadership in Artificial Intelligence* (Jan. 23, 2025), <https://www.whitehouse.gov/presidential-actions/2025/01/removing-barriers-to-american-leadership-in-artificial-intelligence/>.



commitment to connecting all Americans and ensuring strong U.S. supply of optical fiber and cable, Corning operates five cable plants and two fiber plants in North Carolina. We have invested more than \$500 million in fiber and cable manufacturing since 2020, nearly doubling our ability to serve the U.S. cable market. Our focus remains on championing U.S. manufacturing and job creation.

As the NITRD NCO and OSTP are aware, fiber has become the de-facto transmission media across the nation's data center infrastructure thanks to its unequalled characteristics: high speed, low latency, ultra-high-density, near limitless capacity, security, sustainability, and high durability/reliability. As a future-proof technology, fiber will also remain a critical component of the high-speed broadband telecommunications infrastructure that data centers rely on for decades to come.

These comments seek to provide the NITRD NCO and OSTP with insight into the role that Corning plays in fueling AI and the data center industry, as well as recommendations for the development of the AI Action Plan to help foster Federal policies that ensure that our nation's data centers keep pace with growing demand.

II. CORNING SUPPLIES ESSENTIAL PRODUCTS INCLUDING OPTICAL FIBER, FIBER OPTIC CABLE, AND FIBER OPTIC CONNECTIVITY SOLUTIONS TO THE HYPERSCALE DATA CENTERS THAT FUEL AI AND OUR CONNECTED WORLD.

Data centers have long relied on fiber broadband connections to carry out their operations, as fiber is the most capable, reliable, scalable, multi-use broadband infrastructure available. With its symmetrical speeds, near infinite bandwidth, and unmatched reliability, fiber ensures that broadband infrastructure is “future-proof.” Optical fiber is also the only technology that does not impose an intrinsic bottleneck on the transmission medium.

Optical fiber networks have revolutionized data transmission, and as the inventor of the first low-loss optical fiber, Corning has been a leader in this revolution. Today, there are approximately eight billion kilometers of fiber deployed around the globe—enough to travel to the sun 54 times.² Simply put, fiber—and Corning fiber, in particular—has connected Americans across the country with resilient, fast, and trusted broadband service, and today's high-speed Internet, voice, and video connections would not be possible without Corning's innovations in optical fiber. The same holds true for Corning's role with data centers in the United States.

According to a recent report by Synergy Research Group (“Synergy”), “the number of large data centers operated by hyperscale providers increased to 992 at the end of 2023, and passed the thousand mark in early 2024.”³ In terms of future growth, Synergy found that it took just four years for the total capacity of hyperscale data centers to double, and that this capacity

² See *The Evolution of Optical Fiber*, Corning, <https://www.corning.com/media/worldwide/coc/documents/Fiber/evolution-of-optical-fiber.pdf> (last visited Oct. 31, 2024).

³ See *Hyperscale Data Centers Hit the Thousand Mark; Total Capacity is Doubling Every Four Years*, Synergy Research Group (Apr. 17, 2024), <https://www.srgresearch.com/articles/hyperscale-data-centers-hit-the-thousand-mark-total-capacity-is-doubling-every-four-years>.



will double again in the next four years.⁴ Synergy also expects that 120-130 additional hyperscale data centers will come online each year, and “capacity growth will be driven increasingly by the even larger scale of those newly opened data centers, with generative AI technology being a prime reason for that increased scale.”⁵

The demand for artificial intelligence changes the game for data centers. Emerging AI data center network architectures represent a step change in passive optical content in the data center—and at Corning, we’re excited to be working with the world’s largest hyperscale data centers to lay the fiber-rich foundation required for AI data center operations. To support current and future data flow demands, hyperscalers require modern fiber optic connectivity, and Corning has a long history of innovation to meet hyperscalers’ demands. Corning’s fiber, cable, and connectivity products are used extensively inside of the data center to connect processors and servers, within the data center campus to connect data halls, and in long-haul networks to connect data centers nationwide. And just as importantly, we see this trend being a game-changer in the access network. To use this powerful compute capability, it requires high bandwidth networks that enterprises and consumers can access to realize AI’s potential.

As critical and emerging technologies like AI accelerate demands for more computing infrastructure, hyperscalers will no doubt be faced with unique challenges. For example, inside the data center, the new large Graphical Processing Unit, or “GPU,” clusters that AI calls for, require 10 times more the fiber connections than traditional data centers.⁶ Hyperscalers must be prepared to implement the smallest, densest fibers and connectors, as large language models, for example, require an enormous amount of fiber connectivity. These denser connections require product innovations across fiber, cable, and connectivity, and Corning is investing heavily in those capabilities in the United States.

Corning has focused on the vectors of size, speed, and simplicity to create our next generation of fiber and cable solutions to help operators manage scalable network growth. Corning is seeing strong adoption of these solutions, which is helping hyperscale operators get these GPU clusters up and running faster, while also increasing the density of the network. We have reinvented our fiber to be 40% smaller than legacy fibers, allowing for tighter bends and greater flexibility in deployments. This smaller fiber paves the way for a revolution in cable density—doubling the fiber count in the same cable diameter. For our customers, that reduces the need for additional infrastructure buildouts. As highlighted in an article from Light Reading in July 2024, it was noted that “Blackwell GPU [from Nvidia] rack houses 72 GPUs, each connected by 8 fibers, for a total of 576 fibers, or 18X as many fibers as a legacy CPU rack.”⁷

⁴ See *Id.*

⁵ *Id.*

⁶ See Joe Gillard, *A Q&A With Corning’s Michael Crook on Data Center Growth*, ISE Magazine (Sept. 26, 2024), <https://www.isemag.com/cande-netdev-ops-gis-open-source-networks/data-centers/article/55143085/a-qa-with-corning-michael-crook-on-data-center-growth> (“ISE Magazine Interview”).

⁷ See Mike Dano, *Corning lifts expectations amid fiber demand in AI data centers*, Light Reading (July 8, 2024), <https://www.lightreading.com/data-centers/corning-lifts-expectations-amid-fiber-demand-in-ai-data-centers>.



As the creator and largest producer of fiber optics in the world, Corning stands ready to provide hyperscalers and network providers with the infrastructure they need to support the future growth of AI and hyperscale data center solutions. Thanks to Corning's trusted relationships with the world's largest hyperscalers and its commitment to research and development, Corning is well suited to anticipate unforeseen challenges that hyperscalers may encounter and to innovate proactively to address them.

III. U.S. AND NORTH AMERICAN COMMUNICATIONS INFRASTRUCTURE MANUFACTURING IS CRITICAL TO NATIONAL SECURITY.

With respect to AI and the communication infrastructure eco-system, the North American fiber industry includes fiber and cable manufacturing within the United States, as well as connectivity product manufacturing by U.S. companies in Mexico. Moreover, given the multitude of individual components necessary for enabling data centers to operate on a day-to-basis (e.g., fiber, chips, networking equipment, and more), it is critical that the Federal government protect and support U.S. manufacturers by ensuring the resiliency of the supply chain for these products. The U.S. Department of Commerce ("DOC") and the U.S. Department of Homeland Security ("DHS") have previously worked together to evaluate and protect the U.S. Information and Communications Technology ("ICT") industry which includes fiber optics and other connectivity products from supply chain vulnerabilities that were caused by the COVID-19 pandemic in a report released in February 2022. The report states, "The U.S. ICT industry relies on globalized and complex supply chains which complicates industry's ability to elucidate all suppliers and ensure product integrity and security throughout the supply chain. The lack of supply chain transparency and security assurance presents several risks, such as the insertion of counterfeit or used parts into critical hardware components and the injection of malicious software code. While the private sector must take the lead on building more transparency and security into their supply chains, the U.S. Government should promote such practices through the following actions including promoting supply chain risk management practices through procurement and monitoring efforts."⁸

As the Fiber Broadband Association also highlighted in their "Trusted Fiber" white paper published in February 2024, "national security must be a top consideration in the deployment of critical infrastructure. Trusting our fiber sources enables the U.S. to best protect its citizens, economy, and other institutions, notably from foreign aggression or terrorism."⁹ Given the anticipated growth in computing demand from AI and other advanced technologies, and increased demand for data centers that will result from that growth, Corning agrees that the NITRD NCO and OSTP should ensure that the Federal government takes measures to foster the continued market growth of domestic data centers and to further strengthen supply chain resilience.

⁸ See *Assessment of the Critical Supply Chains Supporting the U.S. Information and Communications Technology Industry*, U.S. Department of Commerce (Feb. 24, 2022), <https://www.commerce.gov/sites/default/files/2022-02/Assessment-Critical-Supply-Chains-Supporting-US-ICT-Industry.pdf>.

⁹ See Fiber Broadband Association, *Recommended Measures to Define and Deploy "Trusted Fiber" in the United States*, (Feb. 27, 2024), https://fiberbroadband.org/wp-content/uploads/2024/03/FBA-Trusted-Fiber_Feb-2023.pdf.



Because of data centers' fiber dependency, continued growth in this industry will necessarily increase demand for fiber, cable, and connectivity products. In anticipation of this demand, as well as the demand spurred by infrastructure funding programs and the domestic "Build America, Buy America" ("BABA") provisions in the Infrastructure Investment and Jobs Act ("IIJA"), Corning has made substantial investments in its manufacturing facilities and capabilities in the United States, including opening a new optical cable manufacturing campus in Hickory, North Carolina in March 2023.¹⁰ Corning maintains a strong and growing domestic manufacturing base in the United States for producing fiber, but there still are opportunities with the AI Action Plan for the United States government to improve data centers' market development, supply chain resilience, and data security.

Corning believes that supply chain resiliency is a critical component of fully connecting Americans in rural and underserved communities. With respect to this RFI specifically, the continued growth of the U.S. data center industry hinges on resilient supply chains and a skilled workforce, among other factors. While the global ICT supply chain is in a better place now than it was during the COVID-19 pandemic, challenges remain. There is sufficient North America capacity for passive optical connectivity products (e.g. frames, racks, cassettes, modules and cable assemblies) for all U.S. data center needs. Just like fiber and cable, Corning believes self-sufficient capacity of passive optical connectivity products in North America is critical to support future AI data center growth in the United States

However, it is important to note that the production of any one of the products that data centers rely upon cannot move forward if there is a delay or shortage of any one component, so the Federal government must be prepared to take action to ensure that the supply chains that feed our data centers remain resilient and run as smoothly as possible.

IV. CORNING FIBER, CABLE, AND CONNECTIVITY SOLUTIONS ALSO PLAY A CRITICAL ROLE IN ESTABLISHING HIGH-SPEED AND RELIABLE NETWORKS THAT CONNECT DATA CENTERS NATIONWIDE.

As detailed above, Corning fiber provides data centers (and hyperscalers in particular) with fast, reliable broadband connections within their facilities. But as data center demand increases, so too will the need for similarly fast and reliable middle mile infrastructure, as robust fiber networks will be needed to interconnect these data hubs. Corning has played a critical role in paving those highways.

Indeed, Corning has been laying the foundation for high-quality middle mile infrastructure, as its fiber has been specifically designed for long distance transmissions. Moreover, Corning has developed relationships with broadband providers across the country to help build out middle mile network infrastructure. In August 2024, Corning and Lumen Technologies ("Lumen") entered into an agreement for a substantial supply of next-generation

¹⁰ See *Corning Opens Optical Cable Manufacturing Campus in North Carolina to Accelerate Broadband Buildouts and Connect the Unconnected*, Corning, News Release (Mar. 29, 2023), <https://www.corning.com/worldwide/en/about-us/news-events/corning-opens-optical-cable-manufacturing-campus-in-north-carolina-to-accelerate-broadband-buildouts-and-connect-the-unconnected.html>.

cable, which will more than double Lumen's U.S. intercity fiber miles.¹¹ Lumen will now be able to offer significant capacity to major cloud data centers that are working to stay ahead of the demands of AI and other high-bandwidth applications. Lumen has since partnered with Meta and Microsoft, two of the largest hyperscalers in the industry, to provide the fiber connectivity that their data centers need as the companies' AI efforts continue to expand.¹²

In addition to the company's partnership with Lumen, Corning recently entered into a multi-year purchase agreement with AT&T to provide next-generation fiber, cable, and connectivity solutions to support the expansion of AT&T's fiber network.¹³ AT&T will rely on Corning fiber to help expand its network through organic investment; its Gigapower joint venture with BlackRock, which is building networks outside of AT&T's footprint; and several recently announced commercial open-access agreements.¹⁴

As demand on data centers grows, so too will these facilities' energy needs. But given the explosive growth in data centers, it will be imperative for industry to scale in a more energy efficient way. Fiber offers the best pathway to achieve this goal. Fiber has played a critical role in helping to reduce the levels of energy being consumed by data centers and will continue to do so moving forward.

For instance, fiber can connect data center campuses to one another, allowing them to transmit between each other at high data rates. This in turn allows for more distributed power access for these data centers, as opposed to requiring them to rely on one location to supply higher levels of power. Also, because fiber optic cables rely on light rather than electric signals, fiber reduces energy consumption by up to 54% compared to copper-based networks,¹⁵ thereby allowing data centers to drastically reduce their energy consumption. Corning, itself, has seen

¹¹ See *Corning and Lumen Reach Supply Agreement on Next-Generation Fiber-Optic Cable to Support Data Center AI Demands*, Corning, News Release (Aug. 1, 2024), <https://www.corning.com/worldwide/en/about-us/news-events/news-releases/2024/08/corning-and-lumen-reach-supply-agreement-on-next-generation-fiber-optic-cable-to-support-data-center-ai-demands.html>.

¹² See Sean Buckley, *Lumen and Meta enter AI network pact*, Lightwave + BTR (Oct. 21, 2024), <https://www.lightwaveonline.com/home/article/55237111/lumen-and-meta-enter-ai-network-pact>.

¹³ See *AT&T and Corning Expand Collaboration with Multi-Year Purchase Agreement*, Corning, News Release (Oct. 28, 2024), <https://www.corning.com/worldwide/en/about-us/news-events/news-releases/2024/10/att-and-corning-expand-collaboration-with-multi-year-purchase-agreement.html>.

¹⁴ See Linda Hardesty, *AT&T CEO John Stankey ponders big-picture thoughts about open access networks*, Fierce Network (Oct. 24, 2024), https://www.fierce-network.com/broadband/att-ceo-john-stankey-ponders-big-picture-thoughts-about-open-access-networks?utm_medium=email&utm_source=nl&utm_campaign=FN-NL-FierceNetworkBroadband&oly_enc_id=8985J6128645H7C.

¹⁵ See *Optical Fiber Sustainability and Safety*, Corning, <https://www.corning.com/optical-communications/worldwide/en/home/products/fiber/optical-fiber-resource-center/environmental-and-safety.html> (last visited Oct. 31, 2024). For additional information on the superior capability of optical fiber to support growing capacity demands, as well as data on its sustainability benefits as the industry scales, please see Corning's white paper, *A sustainable future with optical fiber*, available at <https://www.corning.com/media/worldwide/coc/documents/Fiber/white-paper/WP1000.pdf>.



energy savings of more than 68,000 kWh per year thanks to the deployment of fiber architecture in its optical communications headquarters in Charlotte, North Carolina.¹⁶

V. CONCLUSION

Corning appreciates the opportunity to provide input on the RFI that the NITRD NCO and OSTP issued for public response. As the leader in fiber optic technology, Corning continually seeks to be at the forefront of the latest technology frontier with AI and the rapidly expanding data center industry is no exception. Corning is proud to be playing such a key role in helping data centers expand quickly and responsibly and would welcome the opportunity to further collaborate with the NITRD NCO and OSTP on these issues.

Respectfully submitted,

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¹⁶ See Piers Benjamin, *Using Fiber to the Edge to Cut the Carbon*, dotmagazine (Jan. 2023), <https://www.dotmagazine.online/issues/digital-responsibility-and-sustainability/abundant-energy-through-data-center-waste-heat/fiber-to-edge-to-cut-carbon>.